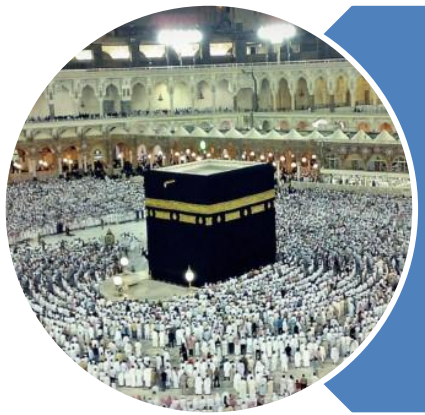
Data Structures - I

CPCS 204

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

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Algorithm Analysis

Introduction

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Algorithm Analysis - **Introduction**

- Efficiency
  - Max # of 1's
    - **O(n)**
  - Linear Search
    - **O(n)**
  - Binary Search
    - **O($\log_2$ n)**
  - Sorted List Matching Problems
    - **O(n)**

Algorithm Analysis - **Introduction**

- We will use Order Notation to approximate two things about algorithms:
  - **How much time they take**
  - How much memory (space) they use
- Note:
  - It is nearly impossible to figure out the exact amount of time an algorithm will take
  - Each algorithm gets translated into smaller and smaller machine instructions
  - Each of these instructions take various amounts of time to execute on different computers

Algorithm Analysis - **Introduction**

- We want to judge algorithms **independent** of their implementation
- Specifically
  - we want to know **how the performance of an algorithm responds to changes in problem size**
- And, rather than figure out an algorithm's exact running time
  - We only want **an approximation**
  - **Big-O** approximation

Algorithm Analysis - **Introduction**

- Big – O Approximation
  - After analyzing an algorithm, we end up with
    - Polynomial equation/function

$$f(n) = 4n^2 + 3n + 10 \quad = O(n^2)$$

  - Growth of the function

n	$4n^2$	3n	10
1	4	3	10
10	400	30	10
100	40,000	300	10
1000	4,000,000	3,000	10
10,000	400,000,000	30,000	10
100,000	40,000,000,000	300,000	10
1,000,000	4,000,000,000,000	3,000,000	10

Algorithm Analysis - **Introduction**

- Big – O Approximation $f(n) = 4n^2 + 3n + 10$
  - Efficiency is approximated by **highest order term**
    - **$4n^2$**
  - Eliminate all **lower order** terms
    - **3n + 10**
  - Ignore all **coefficient constants** too
    - **4** (from $4n^2$)
  - Hence

$$f(n) = 4n^2 + 3n + 10 = O(n^2)$$

Algorithm Analysis - **Introduction**

- Big – O Approximation
  - $f(n) = 4n^2 + 3n + 10$
    - **$O(n^2)$**
  - $f(n) = 76{,}756{,}234n^2 + 427{,}913n + 7$
    - **$O(n^2)$**
  - $f(n) = 74n^8 - 62n^5 - 71562n^3 + 3n^2 - 5$
    - **$O(n^8)$**
  - $f(n) = 42n^4*(12n^6 - 73n^2 + 11)$
    - **$O(n^{10})$**
  - $f(n) = 75n*\log n - 415$
    - **$O(n*\log n)$**

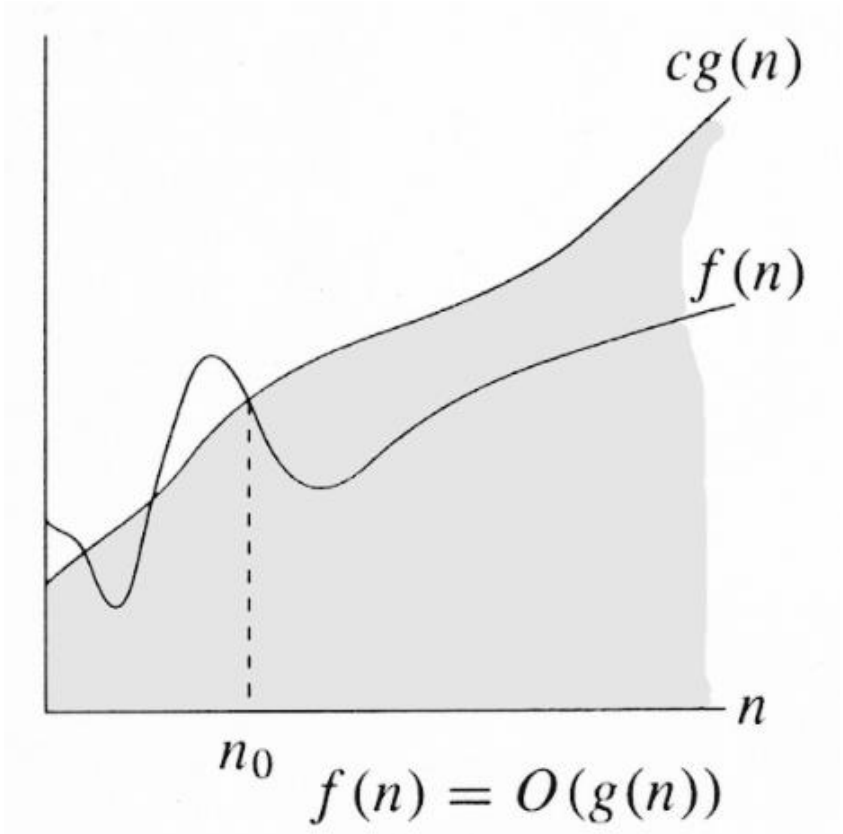
Big–Oh - **Introduction**

- $4n^2 + 3n + 10 = O(n^2)$
- We have one function:
  - **$f(n) = 4n^2 + 3n + 10$**
- Let us make a second function, **g(n)**
  - **$g(n) = n^2$**
- So now we have
  - **f(n) = O (g(n))**

Big–Oh – Introduction (formal definition)

$$O(g(n)) = \{f(n): \text{there exist positive constants } c \text{ and } n_o \text{ such that } 0 \le f(n) \le cg(n) \text{ for all } n \ge n_o\}$$

- c * g(n) is an **upper bound** on the value of f(n)
- Such that
  - **f(n) <= c * g(n)**



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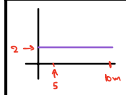
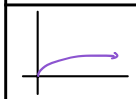
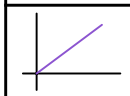
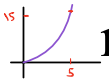
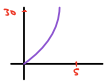
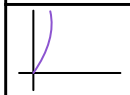
Common Order – Introduction

- Complexity Classes
- The best running time
  - **1**
- The Worst running time
  - **n!**

Function	Name
 1	Constant
 log n	Logarithmic
 n	Linear
 n log n	Poly-log
 n^2	Quadratic
 n^3	Cubic
 2^n	Exponential
 n!	Factorial

Complexity Classes – O(1)

- It is a notation for "constant work"
- An example would be finding the smallest element in a sorted array
  - There's nothing to search for here
  - The smallest element is always at the beginning of a sorted array
  - So this would take O(1) time

Complexity Classes – O(n)

- It means that the **work changes in a way that is proportional to n**
- Example:
  - If the input size doubles, the running time also doubles
- It is a notation for "**work grows at a linear rate**"
- You usually can't really do a lot better than this for most problems we deal with

Complexity Classes – $O(n^k)$

- **$O(n^2)$** or "Order n^2 ": **Quadratic time**
  - If input size doubles, running time increases by a factor of n^2
- **$O(n^3)$** or "Order n^3 ": **Cubic time**
  - If input size doubles, running time increases by a factor of n^3
- **$O(n^k)$**: Other polynomial time
  - Should really try to avoid high order polynomial running times

Complexity Classes – O(Exponential)

- **O(2^n)** or "Order 2^n ": **Exponential time**
  - more theoretical rather than practical interest because they cannot reasonably run on typical computers for even for moderate values of n.
  - Input sizes bigger than 40 or 50 become unmanageable
    - Even on faster computers
- **O(n!)**: even worse than exponential!
  - Input sizes bigger than 10 will take a long time

Complexity Classes – O(Logarithmic)

- **O(n logn)**:
  - Only slightly worse than O(n) time
    - And O(n logn) will be much less than $O(n^2)$
    - This is the running time for the better sorting algorithms we will go over (later)
- **O(log n)** or "Order log n": **Logarithmic time**
  - If input size doubles, running time increases ONLY by a constant amount
  - **any algorithm that halves the data** remaining to be processed on each iteration of a loop will be an **O(log n) algorithm**.

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Algorithm Analysis

Some Example

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Algorithm Analysis - Example - 01

- The number of operations executed by these loops is the sum of the individual loop operations.
- The first loop has **n/2** operations
- The second loop has **n^2** operations
- Not nested, So we simply ADD their operations
- The total number of operations would be
  - **$T(n) = n/2 + n^2$**

```
for (k = 1; k <= n/2; k++) {
        sum = sum + 5;
}
for (j = 1; j <= n*n; j++) {
        delta = delta + 1;
}
```

In Big-O terms, we can express the number of operations as **$O(n^2)$**

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Algorithm Analysis - **Example - 02**

- The outer loop runs **n times**
- for each of those n times, the inner loop also runs **n times**
- So we have **n operations per iteration** of OUTER loop
- Finally, we have the number of **operations**
  - **T(n) = n*n**

```
int func1(int n) {
        int i, j, x = 0;
        for (i = 1; i <= n; i++) {
                for (j = 1; j <= n; j++) {
                        x++;
                }
        }
        return x;
}
```

In Big-O terms, we can express the number of operations as **$O(n^2)$**

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Algorithm Analysis - **Example - 03**

- The loops are **NOT nested**
- First loop runs **n times**
- Second loop runs **n times**
- Total runtime is on the order of
  - **T(n) = n+n = 2n**

```
int func3(int n) {
    int i, x = 0;
    for (i = 1; i <= n; i++)
        x++;
    for (i = 1; i<=n; i++)
        x++;
    return x;
}
```

In Big-O terms, we can express the number of operations as **O(n)**

Algorithm Analysis - **Example - 04**

- Initially **n**
- After first iteration
  - **n/2**
- After 2nd iteration
  - **n/4**
- After 3rd iteration
  - **n/8**
- After "k" iteration
  - **$n/2^k$**
- Algorithm ends when
  - **$n/2^k = 1$**
- So **$k = \log_2 n$**

```
int func4(int n) {
        while (n > 0) {
                printf("%d", n%2);
                n = n/2;
        }
}
```

In Big-O terms, we can express the number of operations as **$O(\log_2 n)$**

Algorithm Analysis - Example - 05

```
int func5(int[][] array, int n)  {
        int i = 0, j = 0;
        while (i < n) {
                while (j < n && array[i][j] == 1)
                        j++;
                i++;
        }
        return j;
}
```

- Outer loop can never run more than **n times**
- Inner loop can never run more than **n times**
- The MOST number of times these two statements can run (combined) is **2n times**
- we approximate the running time as **O(n)**

Algorithm Analysis - Example - 06

```
int func6(int[][] array, int n)  {
        int i = 0, j;
        while (i < n) {
                j = 0;
                while (j < n && array[i][j] == 1)
                        j++;
                i++;
        }
        return j;
}
```

- Outer loop will run **n times**
- Each outer iteration, Inner loop will run **n times**
- The MOST number of times these two statements can run **n*n times**
- we approximate the running time as **$O(n^2)$**

Algorithm Analysis - Example - 07

```
int func7(int A[], int sizeA, int B[], int sizeB) {
        int i, j;
        for (i = 0; i < sizeA; i++)
                for (j = 0; j < sizeB; j++)
                        if (A[i] == B[j])
                                return 1;
        return 0;
}
```

- the runtime here is **NOT in terms of n**
- The outer loop runs **sizeA times**
- For each iteration of outer loop, the inner loop runs **sizeB times**
- So this algorithm runs **sizeA*sizeB** times
- We approximate the running time as **O(sizeA*sizeB)**

Algorithm Analysis - Example - 08

```
int func8(int A[], int sizeA, int B[], int sizeB) {
        int i, j;
        for (i = 0; i < sizeA; i++) {
                if (binSearch(B, sizeB, A[i]))
                        return 1;
        }
        return 0;
}
```

- the runtime here is **NOT in terms of n**
- The outer loop runs **sizeA times**
- For each iteration of outer loop, the inner loop runs lg(sizeB) times
- So this algorithm runs **sizeA*lg (sizeB)** times
- We approximate the running time as **O(sizeA*lg(sizeB))**

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Algorithm Analysis

Practical Problems

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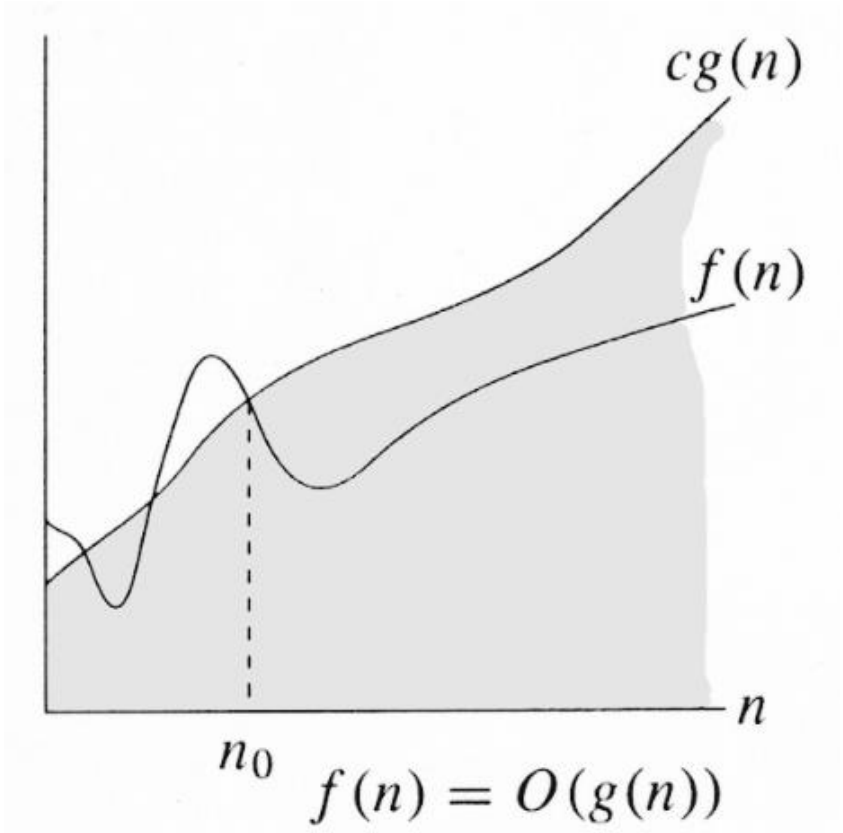
Big–Oh – Introduction (formal definition)

$$O(g(n)) = \{f(n): \text{there exist positive constants } c \text{ and } n_o \text{ such that } 0 \le f(n) \le cg(n) \text{ for all } n \ge n_o\}$$

- c * g(n) is an **upper bound** on the value of f(n)
- Such that
  - **f(n) <= c * g(n)**



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Practical Problems – Example 01

- An algorithm **A** runs in **O(n)** time
- For input size of **10**, the algorithm takes **2** milliseconds
- How much time it will take for input size of **500**?

- T(n) = O (n)
  - **T(n) = c * n**
- n = 10, T(10) = 2ms
  - T(10) = c * 10
  - 2 = c * 10
  - c = 1/5
- c = 1/5, n = 500, T(500)=?
  - T(500) = (1/5) * 500
  - T(500) = 100 ms

Practical Problems – **Example 02**

- An algorithm **A** runs in **$O(n^2)$** time
- For input size of **4**, the algorithm takes **10** milliseconds
- How much time it will take for input size of **16**?

- $T(n) = O(n^2)$
  - **$T(n) = c * n^2$**
- n = 4, T(4) = 10ms
  - $T(4) = c * 4^2$
  - $10 = c * 16$
  - $c = 10/16$
- c = 10/16, n = 16, T(16)=?
  - $T(16) = (10/16) * 16^2$
  - T(16) = 160 ms

Practical Problems – Example 03

- An algorithm **A** runs in **$O(\log_2 n)$** time
- For input size of **16**, the algorithm takes **28** milliseconds
- How much time it will take for input size of **64**?

- $T(n) = O(\log_2 n)$
  - **$T(n) = c * \log_2 n$**
- n = 16, T(16) = 28ms
  - $T(16) = c * \log_2 16$
  - $28 = c * 4$
  - $c = 7$
- c = 7, n = 64, T(64)=?
  - $T(64) = 7 * \log_2 64$
  - $T(64) = 7 * 6$
  - $T(64) = 42$ ms

Practical Problems – Example 04

- An algorithm **D** runs in **$O(\log_2 n)$** time
- For input size of **8**, the algorithm takes **2** milliseconds
- How much time it will take for input size of **64**?

- $T(n) = O(\log_2 n)$
  - **$T(n) = c * \log_2 n$**
- n = 8, T(8) = 2ms
  - $T(8) = c * \log_2 8$
  - $2 = c * 3$
  - $c = 2/3$
- c = 2/3, n = 64, T(64)=?
  - $T(64) = 2/3 * \log_2 64$
  - $T(64) = 2/3 * 6$
  - T(64) = 4 ms

Practical Problems – Example 05

- An algorithm **A** runs in **$O(n^2)$** time
- For input size of **10**, the algorithm takes **30** milliseconds
- What will be the problem size if it takes **120**ms to execute?

- $T(n) = O(n^2)$
  - **$T(n) = c * n^2$**
- n = 10, T(10) = 30ms
  - $T(10) = c * 10^2$
  - $30 = c * 100$
  - $c = 30/100 = 3/10$
- c = 3/10, n = ?, T(n)=120
- $120 = (3/10)*n^2$
- $n^2 = 1200/3 = 400$
- n = 20

Data Structures - I

"No Bonus" marks for this course anymore

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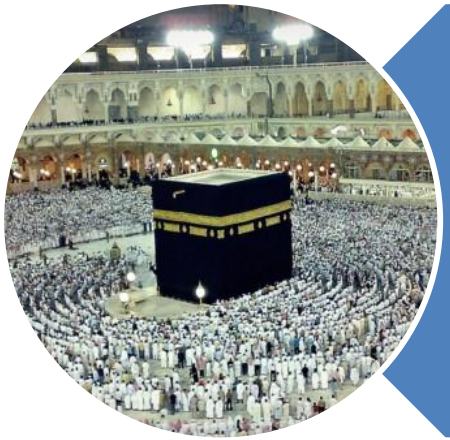

Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us).(Quran 3 : 173)

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